

Usability Testing & Analysis Report

Project Title

Usability Testing and Analysis of an E-Commerce Shopping Application

1. Introduction

Usability testing is a user-centered evaluation method used to measure how easily users can interact with a digital product. The purpose of this study was to identify usability problems, evaluate user satisfaction, and recommend design improvements for an e-commerce shopping application.

The testing focused on common shopping activities such as user registration, product search, adding products to the cart, completing checkout, and tracking orders.

2. Project Objective

The objectives of this usability testing were to:

- Evaluate the overall user experience.
 - Measure task completion success.
 - Identify usability issues.
 - Collect user feedback.
 - Recommend design improvements.
 - Improve efficiency, effectiveness, and user satisfaction.
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3. Product Description

The tested application is an online shopping platform that allows users to:

- Register and log in
 - Browse products
 - Search products
 - Filter products
 - Add items to cart
 - Complete purchases
 - Track orders
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4. Testing Method

Testing Type: Moderated Usability Testing

Method: Think-Aloud Protocol

Environment: Remote Testing

Duration: 30 Minutes per Participant

Number of Participants: 5

5. Participant Information

Participant	Age	Experience Level
User 1	20	Beginner
User 2	24	Intermediate
User 3	28	Intermediate
User 4	34	Advanced
User 5	40	Beginner

6. Test Scenarios

Scenario 1

A new customer wants to purchase wireless headphones.

Tasks

- Create an account
 - Login
 - Search for headphones
 - Open product details
 - Add to cart
 - Checkout
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Scenario 2

A returning customer wants to reorder a previous purchase.

Tasks

- Login
 - View previous orders
 - Reorder product
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Scenario 3

A customer wants to track an existing order.

Tasks

- Login
 - Open Orders
 - View Tracking Status
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7. Test Tasks

Task	Goal
Register	Create new account
Login	Access account
Search Product	Find product
View Product	Read product details
Add to Cart	Add selected item
Checkout	Complete purchase
Track Order	View delivery status

8. Usability Metrics

The following metrics were used during testing:

- Task Success Rate
 - Time on Task
 - Error Rate
 - User Satisfaction
 - Navigation Efficiency
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9. Test Results

Task	Success Rate	Average Time
Register	100%	1 min
Login	100%	30 sec
Search Product	80%	2 min
View Product	100%	1 min
Add to Cart	80%	1 min
Checkout	60%	4 min
Track Order	100%	45 sec

10. Observations

Positive Findings

- Registration process was simple.
- Product pages were visually attractive.
- Search feature worked effectively.
- Order tracking was easy to use.

Negative Findings

- Checkout button was difficult to locate.
 - Product filters were confusing.
 - Font size was too small.
 - Cart icon lacked visibility.
 - Payment page contained too much information.
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11. User Feedback

Participant comments included:

- "Checkout should be simpler."
 - "Buttons should be larger."
 - "Improve filter organization."
 - "Increase font size."
 - "Use stronger color contrast."
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12. Usability Issues

High Priority

- Checkout process too complex
- Poor CTA visibility
- Confusing payment screen

Medium Priority

- Product filter organization
- Small touch targets

Low Priority

- Minor typography inconsistencies
 - Icon spacing improvements
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13. Recommendations

Navigation

- Improve menu visibility.
- Highlight important actions.

Checkout

- Reduce the number of checkout steps.
- Make the "Place Order" button larger and more prominent.

Product Filters

- Group filters by category.
- Add clear labels.

Typography

- Increase font size.
- Improve readability.

Accessibility

- Increase color contrast.
 - Add descriptive labels for icons.
 - Ensure adequate touch target sizes.
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14. Conclusion

The usability study demonstrated that users could successfully complete most primary tasks. However, the checkout flow and product filtering process introduced unnecessary friction and confusion.

By simplifying navigation, improving visual hierarchy, increasing accessibility, and streamlining the checkout process, the application can significantly enhance usability, reduce user errors, and improve overall customer satisfaction.

15. Expected Learning Outcome

This project demonstrates practical understanding of:

- User-centered design
- Usability testing methods
- User observation
- UX research
- Data analysis
- Problem identification
- Design iteration
- UX reporting
- Continuous product improvement